



Medical Insurance Data Science Dashboard

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This dashboard performs exploratory data analysis, statistical hypothesis testing, and multiple linear regression on the medical insurance dataset.

Dataset Preview

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.9	0	yes	southwest	16884.924
1	18	male	33.77	1	no	southeast	1725.5523
2	28	male	33	3	no	southeast	4449.462
3	33	male	22.705	0	no	northwest	21984.4706
4	32	male	28.88	0	no	northwest	3866.8552

Data Exploration Hypothesis Testing Lab Live Prediction & Diagnostics

Data Exploration

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Dataset Information

Number of Records

1338

Number of Variables

7

Missing Values

0

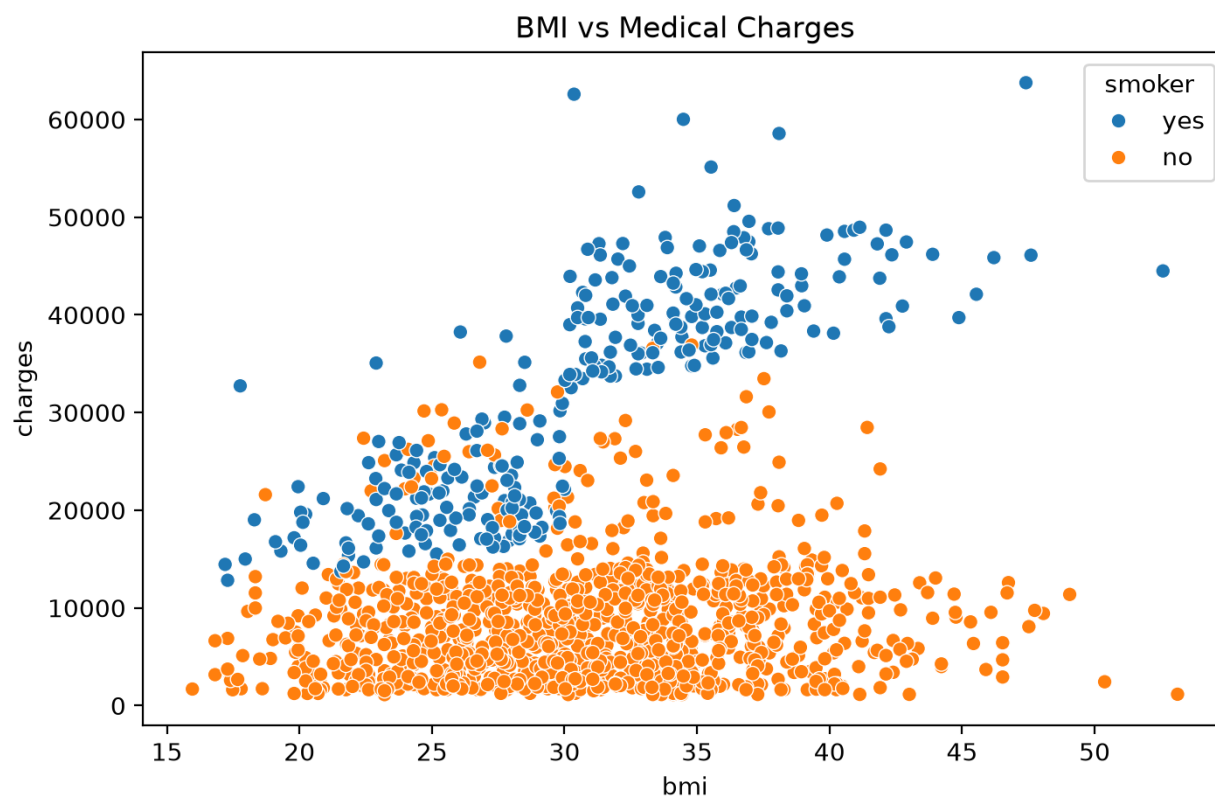
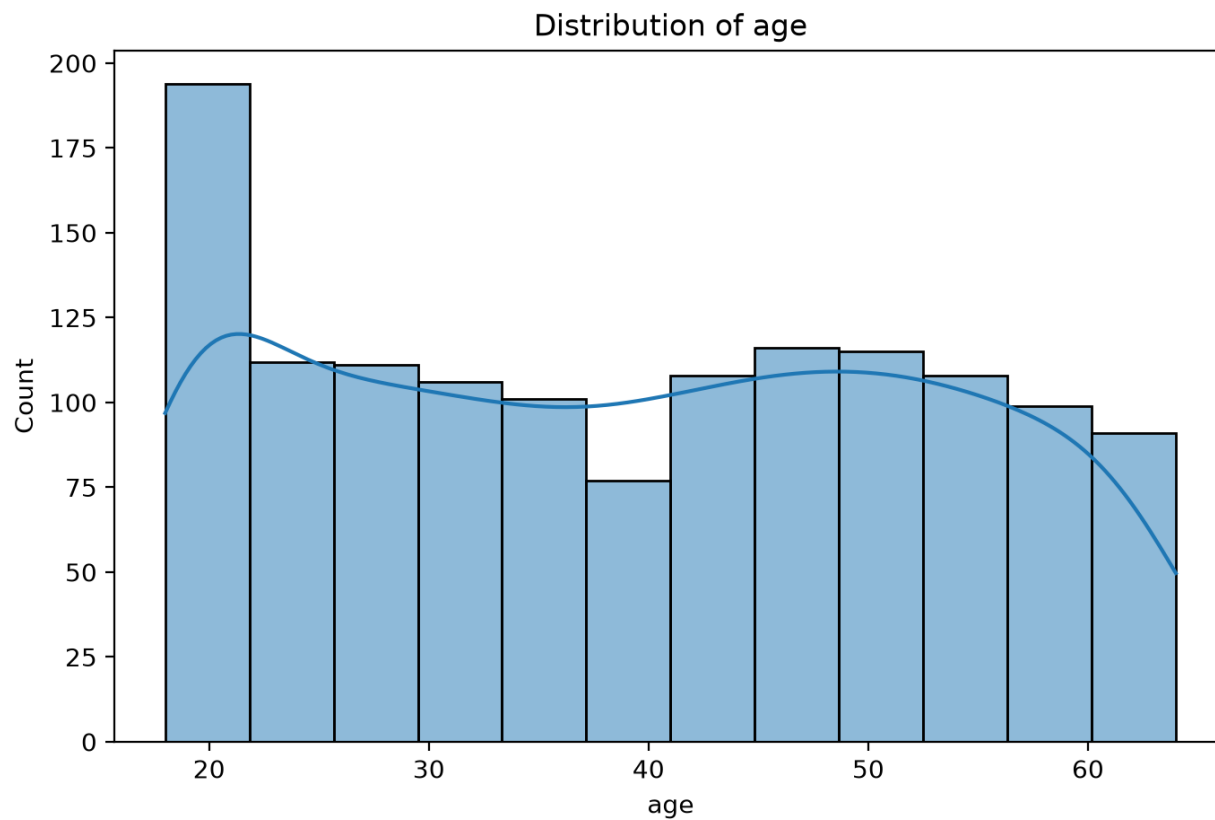
Descriptive Statistics

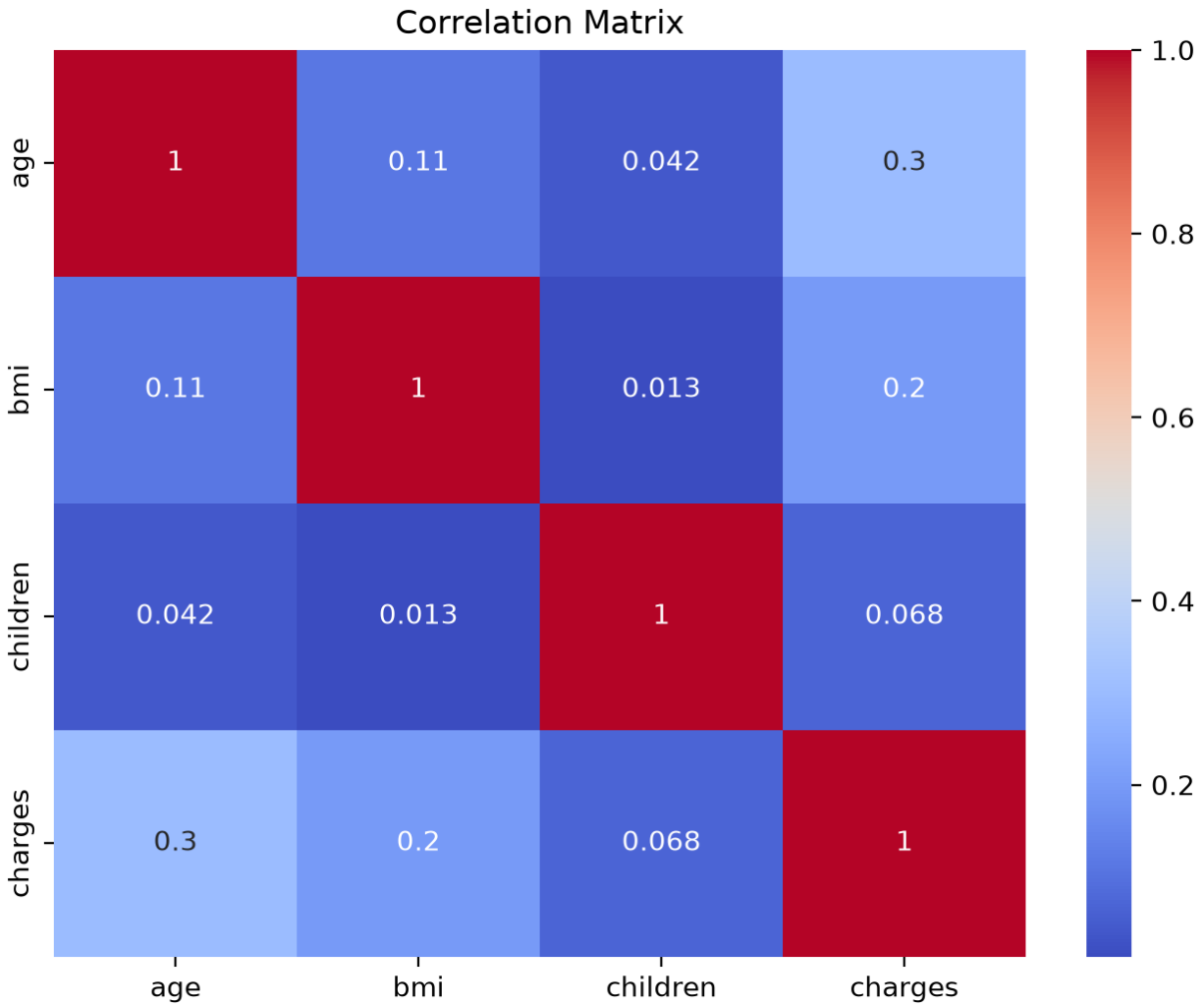
Summary Statistics

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	age	bmi	children	charges
count	1338	1338	1338	1338
mean	39.207	30.6634	1.0949	13270.4223
std	14.05	6.0982	1.2055	12110.0112
min	18	15.96	0	1121.8739
25%	27	26.2963	0	4740.2872
50%	39	30.4	1	9382.033
75%	51	34.6938	2	164

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Hypothesis Testing Lab

Two-Group Comparison

This section compares medical charges between two groups. The Shapiro-Wilk test checks normality, and Levene's test checks equality of variances. Depending on the assumptions, either an independent-samples t-test or Mann-Whitney U test is used.

Select grouping variable

smoker

Select numerical variable

charges

Run Two-Group Test

Test Result

```
{
  "groups": "(ArrowStringArray) of 'yes' (no) \length: 2, dtype: str"
```

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Test Result

▼

{

"groups" : "<ArrowStringArray>\n['yes', 'no']\nLength: 2, dtype: str"

▼

"shapiro_group1" : [

0 : 0.9395519382929163

1 : 3.624989722535372e-9

]

▼

"shapiro_group2" : [

0 : 0.8728637657660203

1 : 1.4459006863640269e-28

]

▼

"levene" : [

0 : 332.6135162726081 🚩

1 : 1.5593284881791394e-66

]

▼

"test_name" : "Mann-Whitney U Test"

▼

"test_result" : [

0 : 284133

]

}

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One-Way ANOVA

One-Way ANOVA evaluates whether the mean medical charges differ significantly across the four geographical regions.

Run ANOVA

▼

{

"groups" : "<ArrowStringArray>\n['southwest', 'southeast', 'northwest', 'northeast']\nLength: 4, dtype: str"

▼

"anova_result" : [

0 : 2.96962669358912

1 : 0.03089335607049739

]

▼

"conclusion" : "Reject H0"

}

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Live Prediction & Diagnostics

Multiple Linear Regression Model

The regression model predicts medical insurance charges using age, BMI, number of children, smoking status, and geographical region.

Regression Summary

Model Information

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	0	1	2	3
0 Dep. Variable:	charges	R-squared:	0.751	
1 Model:	OLS	Adj. R-squared:	0.750	
2 Method:	Least Squares	F-statistic:	572.7	
3 Date:	Wed, 09 Sep 2026	Prob (F-statistic):	0.00	
4 Time:	16:11:09	Log-Likelihood:	-13548.	
5 No. Observations:	1338	AIC:	2.711e+04	
6 Df Residuals:	1330	BIC:	2.715e+04	

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Estimated Coefficients

		coef	std err	t	P> t	[0.025	0.975]
0	const	-1.199e+04	978.762	-12.250	0.000	-1.39e+04	-1.01e+04
1	age	256.9736	11.891	21.610	0.000	233.646	280.301
2	bmi	338.6646	28.559	11.858	0.000	282.639	394.690
3	children	474.5665	137.740	3.445	0.001	204.355	744.778
4	smoker_yes	2.384e+04	411.856	57.875	0.000	2.3e+04	2.46e+04
5	region_northwest	-352.1821	476.120	-0.740	0.460	-1286.211	581.847
6	region_southeast	-1034.3601	478.537	-2.162	0.031	-1973.130	-95.590
7	region_southwest	-959.3747	477.778	-2.008	0.045	-1896.656	-22.094

Regression Coefficient Details

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Model Performance

	Metric	Value
0	R-squared	0.7509
1	Adjusted R-squared	0.7496
2	F-statistic	572.6965
3	F-statistic p-value	0
4	Number of Observations	1338

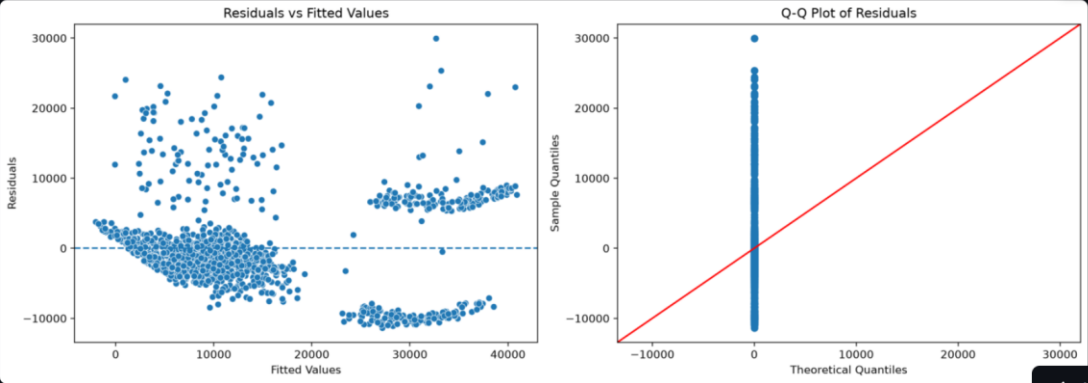
Variance Inflation Factor

	Variable	VIF
0	age	1.0162
1	bmi	1.1042
2	children	1.0037
3	smoker_yes	

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Regression Diagnostics

These diagnostic plots are used to evaluate the assumptions of the multiple linear regression model.



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Age

57- +

BMI

15.00- +

Number of Children

5- +

Smoker

no ▾

Region

northwest ▾

Predict Medical Charges

Predicted Medical Charges: \$22,100.30

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